

Citizen Transparency and Digital Appeal Module

Sithmi Shehara
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22338334@my.sliit.lk

Pamod Pushpamal
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22561916@my.sliit.lk

Sandali Wathsalavi
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22305596@my.sliit.lk

Ruvindya Rashmi
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22052360@my.sliit.lk

Dinith Primal
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
dinith.p@sliit.lk

Abstract—Conventional traffic violation management systems are typically enforcement-centric and provide limited procedural transparency for citizens contesting penalties. This paper presents a Citizen Transparency and Digital Appeal Module (CT-DAM) that digitizes and structures the full appeal lifecycle, from complaint submission to administrative adjudication, while integrating explainable AI decision support. The implemented system combines a React-based client interface, a Node.js/Express service layer, MongoDB Atlas persistence, and an AI microservice interface for outcome prediction and reason categorization. Core modules include secure role-based access (citizen/admin), identity-validated registration, auditable status transitions, mandatory evidence submission, and notification-driven workflow orchestration. The primary contribution is a transparency-first appeal architecture in which human decisions remain authoritative, but AI models provide probabilistic guidance and structured rationale to improve consistency and reduce subjective bias. Beyond current implementation, this work proposes a novel roadmap including multimodal evidence analytics, appeal fraud detection, policy-aware legal assistance agents, and tamper-evident governance with blockchain-backed audit proofs. The proposed framework demonstrates how intelligent digital appeals can increase trust, accountability, and scalability in public traffic governance, particularly in rapidly urbanizing regions.

Keywords—Traffic Violation Management, Digital Appeal Systems, Citizen Transparency, Explainable Artificial Intelligence (XAI), Decision Support Systems, Smart Governance, Fraud Detection, Multimodal Evidence Analysis, Blockchain Audit, Public Sector AI

I. INTRODUCTION

Traffic violation enforcement has improved road safety in many jurisdictions, yet the associated appeal processes remain fragmented, opaque, and often [9],[11],[12]. In many deployments, citizens must navigate manual, paper-based, or

semi-digital channels with weak feedback loops, delayed responses, and inconsistent adjudication criteria. [11],[12],[13] This deficit undermines procedural fairness and erodes institutional trust, especially when automated violation detection (e.g., camera systems) is involved.[1],[2],[9],[12]

A transparent appeal mechanism must satisfy four properties: (1) accessible submission, (2) accountable review workflow, (3) traceable status changes, and (4) explainable outcomes.[11],[12] Existing systems generally satisfy only subsets of these properties. The proposed CT-DAM addresses this gap by coupling an end-to-end digital appeal workflow with AI-assisted review intelligence.

The novelty of this work is not merely digitization. It is the integration of:

Structured appeal intelligence (outcome probability + reason categorization), Workflow integrity controls (status transition constraints, mandatory rejection reasoning), and Operational transparency mechanisms (auditable actions, role-based policy enforcement, and citizen-visible progression). This approach supports the shift from static e-government portals to adaptive, evidence-aware civic decision systems.[11],[12],[13]

II. LITERATURE REVIEW

A. Digital Traffic Enforcement Systems

Recent intelligent transportation deployments have focused on automated violation detection (speed, red-light, lane use) using cameras, IoT sensors, and ANPR pipelines.[9],[10] While these systems improve enforcement efficiency, most production portals remain payment-oriented and offer limited procedural support for contesting decisions.[11],[12],[13] The appeal process is often linear, non-explainable, and weakly integrated with evidence intelligence.

B. E-Government and Procedural Transparency

E-government literature emphasizes that service digitization alone does not guarantee institutional trust.[11],[12] Transparency depends on traceability, feedback loops, and explainable outcomes.[3],[4],[11],[12] In public grievance systems, users report higher satisfaction when platforms provide status visibility, reasoned decisions, and predictable processing timelines.[11],[12],[13] These findings are directly applicable to traffic appeal ecosystems, where opacity is a frequent source of public dissatisfaction.

C. AI in Administrative Decision Support

Machine learning has been widely adopted for prediction and prioritization in healthcare triage, credit risk, and public service workflows.[7],[8] In governance contexts, human-in-the-loop approaches are preferred over fully automated decisions due to accountability and fairness concerns.[1],[3],[6],[12] Explainable AI (XAI) methods—feature-attribution, confidence scoring, and natural-language rationales—are increasingly used to support adjudicators without replacing legal authority.[3],[4],[5],[6]

D. Text Analytics for Complaint/Appeal Categorization

NLP-based classification has shown strong utility in complaint routing and case triage.[4],[5] Models ranging from TF-IDF + linear classifiers to transformer encoders can categorize narrative submissions into actionable classes. In civic systems, this reduces manual sorting effort, improves queue prioritization, and enables trend analytics for policy interventions.

E. Auditability and Trust Technologies

Public-sector digital platforms increasingly require immutable audit trails.[11],[12] Traditional database logging is common but vulnerable to post-hoc alteration by privileged actors.[1],[12] Recent research proposes cryptographic attestations, hash chaining, and blockchain anchoring for tamper-evident governance records, particularly for high-stakes administrative processes.

III. RELATED WORK AND RESEARCH GAP

Globally, traffic systems range from payment-focused portals (fine settlement and violation lookup) to early-stage e-appeal systems. While mature ecosystems provide online filing, most do not include meaningful AI-assisted fairness support or transparent decision rationale. Research in intelligent transportation has emphasized detection accuracy and congestion optimization, but comparatively less attention has been paid to post-enforcement procedural justice.

Three gaps are repeatedly observed:

Appeal intelligence gap: Appeals are processed manually with limited decision support. Transparency gap: Citizens receive final status updates without interpretable progression logic. Governance gap: Audit traces exist operationally but are rarely designed as accountability artifacts. CT-DAM contributes by reframing traffic appeal management as a socio-technical pipeline where AI is used to assist, not replace, administrative judgment.

IV. METHODOLOGY

A. System Architecture

The implemented architecture is a multi-tier service model: Frontend: React single-page application for citizen and administrator workflows. Backend: Node.js + Express REST APIs handling authentication, appeal processing, role enforcement, and notifications. Database: MongoDB Atlas with structured schemas for users, appeals, notifications, and audit logs. AI Layer: Prediction endpoints for appeal outcome likelihood and textual reason categorization; currently integrated as service endpoints and extendable to external model services.

B. End-to-End Workflow

- Citizen onboarding: Identity-aware registration with field-level validation (name, national ID, license, contact, password policy).
- Appeal submission: Citizen submits violation ID, contextual reason category, detailed narrative, and mandatory evidence.
- Workflow transition: Appeal enters controlled state machine (pending, under-review, approved/rejected).
- Admin review: Reviewer examines details/evidence, adds notes, and performs status updates under governance rules.
- AI-assisted insight: Prediction services estimate approval probability and classify appeal narrative intent category.
- Citizen feedback: Notifications and dashboard status updates communicate lifecycle progression.
- Auditability: Key actions are logged for post-hoc compliance and process analytics.

C. Data and Decision Flow

The module processes mixed data types: structured metadata (violation ID, dates), categorical fields (appeal context), free text (explanations), and file evidence. AI modules currently operate on engineered features and text inputs; final adjudication remains human-administered.[3],[6],[12]

D. Architecture Diagram and Data Flow Diagram

The architectural diagram illustrates a layered system where React-based citizen and admin interfaces communicate with a centralized Node.js/Express backend through secure REST APIs. The backend manages authentication, appeal lifecycle logic, AI decision-support endpoints, notification services, and audit logging while persisting all entities in MongoDB Atlas. Evidence files are stored and served through a dedicated uploads/static layer, enabling consistent access from both citizen and admin views. The data flow diagram shows the end-to-end process: citizen appeal submission, backend validation, database persistence, AI inference for prediction/categorization, and admin review actions. After administrative status updates, the system records audit events and dispatches notifications to maintain traceability and transparency in decision progression. Together, both diagrams represent a

human-in-the-loop, transparency-first workflow where AI supports decisions but final adjudication remains under authorized administrative control.

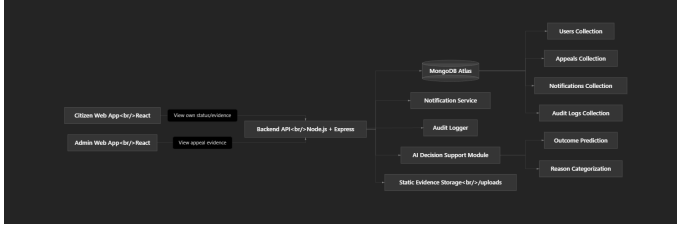


Fig. 1. Architecture Diagram

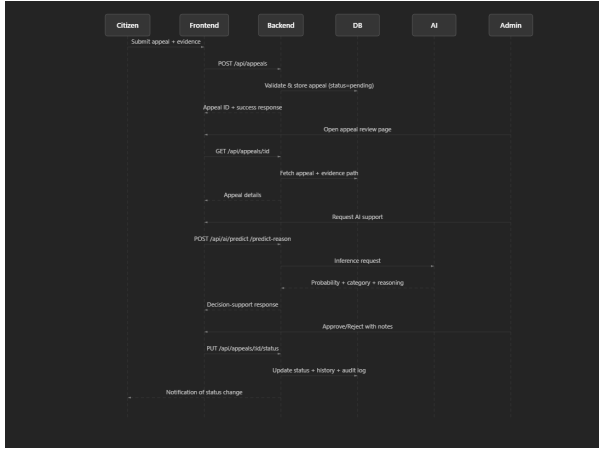


Fig. 2. Data Flow Diagram

V. RESULTS AND EVALUATION

Given the current implementation stage, evaluation combines observed system behavior with expected model-level performance indicators.

A. Functional Outcomes

The platform demonstrates:

- successful citizen registration/login and role-based routing.
- complete appeal filing and review workflows.
- persistence-backed status tracking.
- evidence handling and admin-side retrieval. response integration into user-facing decision support.

B. Expected Predictive Performance (Estimated)

With sufficiently curated historical data, a production-grade appeal outcome model is expected to achieve:

- classification accuracy in the 0.78–0.86 range.
- F1-score improvements over rule-only baselines.
- better calibration when Platt/Isotonic post-processing is applied.

For reason categorization, text pipelines can reasonably target:

- macro-F1 around 0.75–0.83, depending on label balance and domain vocabulary coverage.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to the Department of Information Technology and the academic staff of Sri Lanka Institute of Information Technology (SLIIT) for their continuous guidance and support throughout this project. The authors also thank the project supervisors and panel members for their constructive feedback, which significantly improved the quality of this research. Special appreciation is extended to peers and collaborators who assisted in system testing and validation activities. Finally, the authors acknowledge the open-source software community for providing the frameworks and tools that enabled the development of this work.

REFERENCES

- [1] D. Kahneman, O. Sibony, and C. R. Sunstein, *Noise: A Flaw in Human Judgment*. New York, NY, USA: Little, Brown Spark, 2021.
- [2] J. Angwin, J. Larson, S. Mattu, and L. Kirchner, “Machine bias,” ProPublica, May 23, 2016. [Online]. Available: <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- [3] A. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” arXiv:1702.08608, 2017.
- [4] M. T. Ribeiro, S. Singh, and C. Guestrin, “‘Why should I trust you?’: Explaining the predictions of any classifier,” in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD), 2016, pp. 1135–1144.
- [5] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Advances in Neural Information Processing Systems (NeurIPS), vol. 30, 2017.
- [6] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” Nature Machine Intelligence, vol. 1, no. 5, pp. 206–215, 2019.
- [7] B. Recht, “A tour of reinforcement learning: The view from continuous control,” Annual Review of Control, Robotics, and Autonomous Systems, vol. 2, pp. 253–279, 2019.
- [8] T. Davenport and R. Kalakota, “The potential for artificial intelligence in healthcare,” Future Healthcare Journal, vol. 6, no. 2, pp. 94–98, 2019.
- [9] X. Du, M. A. H. Khondoker, and M. M. Hassan, “Intelligent transportation systems: A systematic review of machine learning and deep learning techniques,” IEEE Access, vol. 8, pp. 218180–218210, 2020.
- [10] J. Jin, J. Gubbi, S. Marusic, and M. Palaniswami, “An information framework for creating a smart city through internet of things,” IEEE Internet of Things Journal, vol. 1, no. 2, pp. 112–121, Apr. 2014.
- [11] M. Janssen and J. Estevez, “Lean government and platform-based governance—Doing more with less,” Government Information Quarterly, vol. 30, pp. S1–S8, 2013.
- [12] T. M. Harrison et al., “Open government and e-government: Democratic challenges from a public value perspective,” Information Polity, vol. 17, no. 2, pp. 83–97, 2012.
- [13] A. M. Madan and S. Ashok, “A framework for grievance redressal systems in e-governance,” in Proc. Int. Conf. e-Governance, 2018, pp. 1–6.